MAT 320 HW10

Aditya Pahuja

Due 11/27

Problem 1. Explain with the ε-δ definition why $f(x) = |x|$ is continuous at $x = 0$ but not differentiable.

Solution. Let $\varepsilon > 0$; then, setting $\delta := \varepsilon$ gives

$$|x - 0| = |x| < \delta = \varepsilon \implies ||x| - |0|| = ||x|| = |x| < \varepsilon,$$

which is what it means to be continuous at 0. On the other hand, for f to be differentiable at zero, we need to check the existence of the limit

$$\lim_{x\to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0} \frac{|x|}{x}.$$

In particular, to show that the limit doesn't exist, it suffices to find two sequences (a_n) and (b_n) that both converge to zero but for which

$$\lim_{n\to\infty} \frac{f(a_n) - f(0)}{a_n - 0} \neq \lim_{n\to\infty} \frac{f(b_n) - f(0)}{b_n - 0}.$$

We do this by taking $a_n = \frac{1}{n}$ and $b_n = -\frac{1}{n}$, since the two limits then evaluate to

$$\lim_{n\to\infty} \frac{|a_n|}{a_n} = \lim_{n\to\infty} \frac{|1/n|}{1/n} = \lim_{n\to\infty} 1 = 1, \quad \lim_{n\to\infty} \frac{|b_n|}{b_n} = \lim_{n\to\infty} \frac{|-1/n|}{-1/n} = \lim_{n\to\infty} -1 = -1,$$

and of course $-1 \neq 1$. Thus the limit that defines the derivative doesn't exist at zero, so f is not differentiable at zero. $\square$

Problem 2. Show that the function $f\colon \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ x^2 \sin(\frac{1}{x}) & \text{otherwise} \end{cases}$$

is differentiable but not of class C^1.

Solution. Clearly the function is differentiable when $x \neq 0$ because x^2, $\sin x$, and $\frac{1}{x}$ are all differentiable functions of x, thus so are their compositions and products, whenever the resulting expression is defined. To check that f is differentiable at zero, we need to show the existence of

$$\lim_{x\to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0} \frac{x^2 \sin(\frac{1}{x})}{x}.$$

This limit is equal to

$$\lim_{x\to 0} x \cdot \sin\left(\frac{1}{x}\right),$$

which is well-known to be 1, so $f'(0)$ exists and is equal to 1.

On the other hand, the derivative when $x \neq 0$ evaluates to

$$2x \cdot \sin\left(\frac{1}{x}\right) - x^2 \cdot \cos\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right) = 2x\sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right).$$

We wish to show that the derivative is not continuous on $\mathbb{R}$, so we will show that the limit of $2x\sin(\frac{1}{x}) - \cos(\frac{1}{x})$ as x approaches zero is not $f'(0) = 1$. Indeed, the sequence $a_n = \frac{1}{2\pi n}$ converges to zero, and

$$\lim_{n\to\infty} 2a_n \sin\left(\frac{1}{a_n}\right) - \cos\left(\frac{1}{a_n}\right) = \lim_{n\to\infty} \frac{2}{2\pi n}\sin(2\pi n) - \cos(2\pi n) = \lim_{n\to\infty} 0 - 1 = -1.$$

Thus, if the limit of $f'(x)$ exists as $x \to 0$, it must equal $-1 \neq f'(0)$, so f' is not a continuous function on $\mathbb{R}$. (In fact the limit doesn't exist at all because the $\cos(1/x)$ term oscillates way too much.) $\square$

Problem 3. Consider the function $f\colon \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Perform the following tasks:

(a) Show that f is differentiable at $x = 0$.

(b) Calculate the derivative function $f'\colon \mathbb{R} \to \mathbb{R}$.

(c) Is f' continuous? Is it differentiable?

Solution. For (a), we need to show the existence of the limit

$$\lim_{x\to 0} \frac{f(x) - f(0)}{x - 0}.$$

For any $\varepsilon > 0$, take $\delta := \varepsilon$. Then, if $|x - 0| < \delta$, we consider two cases:

- If $-\delta < 0 < x$, then $\frac{f(x)-f(0)}{x-0} = 0 < \varepsilon$.
- If $0 \leq x < \delta$, then $\frac{f(x)-f(0)}{x-0} = \frac{x^2}{x} = x < \delta = \varepsilon$.

In either case,

$$|x - 0| < \delta \implies \left|\frac{f(x) - f(0)}{x - 0} - 0\right| < \varepsilon,$$

which is exactly what it means for

$$\lim_{x\to 0} \frac{f(x) - f(0)}{x - 0} = 0,$$

hence the limit exists (and is equal to zero).

We can also compute $f'(x) = 0$ when $x < 0$ because the restriction of f to $(-\infty, 0)$ is just the zero function and $f'(x) = 2x$ when $x > 0$ because the restriction of f to the open interval $(0, \infty)$ is $x \mapsto x^2$. Thus, to answer (b),

$$f'(x) = \begin{cases} 2x & x > 0 \\ 0 & x \leq 0. \end{cases}$$

Finally, we see that $f'(x)$ can be written instead as

$$f'(x) = |x| + x$$

(since when $x > 0$, $|x| + x = x + x = 2x$, while when $x \leq 0$, $|x| + x = -x + x = 0$). This is a sum of continuous functions, hence continuous, but it is not differentiable, since f' being differentiable would mean $f'(x) - x = |x|$ is a sum of differentiable functions, hence differentiable — but we showed in Problem 1 that $|x|$ is not differentiable at zero. □

Problem 5. Evaluate the series

$$\sum_{n=1}^{\infty} \frac{n}{2^n}$$

and

$$\sum_{n=1}^{\infty} \frac{3^n}{n!}.$$

Solution. For the first series, we start with the geometric series

$$\sum_{n=0}^{\infty} x^n,$$

which converges to $\frac{1}{1-x}$ when $|x| < 1$. By the theory we developed in class, the derivative of this function is

$$\sum_{n=1}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2}$$

on the same interval $|x| < 1$. We can then multiply this identity by x to get the identity

$$\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}$$

whenever $|x| < 1$. Since $-1 < \frac{1}{2} < 1$, we can therefore compute

$$\sum_{n=1}^{\infty} n \cdot \frac{1}{2^n} = \frac{1/2}{(1-1/2)^2} = \frac{1/2}{1/4} = \boxed{2}.$$

For the second series, we notice that

$$1 + \sum_{n=1}^{\infty} \frac{3^n}{n!} = \sum_{n=0}^{\infty} \frac{3^n}{n!} = e^3,$$

so the series evaluates to $\boxed{e^3 - 1}$. □

Problem 7. Let f and g be two differentiable functions $\mathbb{R} \to \mathbb{R}$. Show that, given $a < b$, there is a $c \in (a, b)$ such that

$$[f(b) - f(a)] \cdot g'(c) = [g(b) - g(a)] \cdot f'(c).$$

Explain how this statement, Rolle's theorem, and the mean value theorem relate to each other.

Solution. Consider the function $h\colon \mathbb{R} \to \mathbb{R}$ given by

$$h(x) := (f(b) - f(a))g(x) - (g(b) - g(a))f(x).$$

This is a linear combination of differentiable functions, so it is differentiable. Further, we can compute

$$h(a) = h(b) = f(b)g(a) - f(a)g(b),$$

so Rolle's theorem tells us that there is a point in $c \in (a, b)$ at which $h'(c) = 0$, which is exactly what we want.

This theorem is a generalization of the mean value theorem (which is in turn a generalization of Rolle's theorem), since setting $g(x) := x$ for all x spits out the mean value theorem, as there exists a $c \in (a, b)$ such that

$$f(b) - f(a) = (f(b) - f(a)) \cdot g'(c) = (g(b) - g(a)) \cdot f'(c) = (b - a) \cdot f'(c). \qquad \square$$

Problem 8. Let $f\colon [0, 1] \to \mathbb{R}$ be a differentiable function such that $f(0) = 0$ and $f(x) > 0$ for all $x \in (0, 1)$. Prove that there is a number $c \in (0, 1)$ such that

$$\frac{2f'(c)}{f(c)} = \frac{f'(1-c)}{f(1-c)}.$$

Solution. Consider the function $g(x) = f(x)^2 \cdot f(1 - x)$. Its derivative is equal to

$$2f(x) \cdot f'(x) \cdot f(1 - x) - f(x)^2 \cdot f'(1 - x).$$

Moreover, $g(0) = g(1) = 0$, so there is a $c \in (0, 1)$ for which $g'(c) = 0$. Since $f(c) > 0$, we can conclude that

$$0 = \frac{g(c)}{f(c)} 2f'(c)f(1 - c) - f(c)f'(1 - c),$$

which rearranges to

$$2f'(c)f(1 - c) = f(c)f'(1 - c) \iff \frac{2f'(c)}{f(c)} = \frac{f'(1-c)}{f(1-c)}. \qquad \square$$